

Employee Attrition Prediction – Executive Summary

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Project Title: Employee Attrition Prediction using Machine Learning

Tools Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

Dataset: IBM HR Analytics Employee Attrition Dataset

Purpose:

The objective of this project was to identify factors that contribute to employee attrition and develop a predictive model that can help HR teams proactively retain valuable employees.

Key Findings:

The analysis revealed that employee attrition is influenced by a combination of workplace conditions, job responsibilities, and career growth opportunities.

The most significant factors associated with employee turnover were:

- Laboratory Technician job roles
- Overtime work
- Frequent business travel
- Job level and overall work experience
- Long gaps between promotions
- Sales-related positions

Employees who regularly worked overtime or traveled frequently showed a higher likelihood of leaving the organization. Similarly, employees who experienced limited career progression appeared to be at greater risk of attrition.

Model Performance:

Three machine learning models were evaluated: Logistic Regression, Random Forest, and Gradient Boosting.

Among these models, Logistic Regression provided the most effective results for identifying employees who were likely to leave the organization. The model successfully detected a majority of high-risk attrition cases, making it suitable for supporting HR decision-making.

Recommendations:

Based on the findings, the following actions are recommended:

- Monitor and reduce excessive overtime where possible.
- Improve work-life balance through employee wellness initiatives.
- Review business travel policies and provide additional support for frequent travelers.
- Strengthen promotion and career development programs.
- Implement targeted retention strategies for Laboratory Technician and Sales-related roles.
- Utilize predictive analytics to identify high-risk employees and intervene before resignation occurs.

Conclusion:

This project demonstrates that employee attrition can be predicted using historical workforce data. By leveraging these insights, HR teams can make proactive decisions, improve employee satisfaction, reduce turnover costs, and build a more stable and productive workforce.